MIL-F-36938B
8 November 1978
SUPERSEDING
MIL-F-36938A
10 June 1975

MILITARY SPECIFICATION

FORCEPS, HEMOSTATIC, KELLY, 5-1/2 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Kelly hemostatic forceps suitable for accomplishing hemostasis in delicate tissue.

1.2 Classification. The forceps shall be of the following types, as specified (see 6.2).

Type I. - Straight
Type II. - Curved

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.
ASTM E 140	- Hardness Conversion Tables for Metals (Relationship between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General requirements. The forceps shall be in accordance with the applicable requirements of GG-I-526 and as specified herein. In the event of any conflict between the requirements of GG-I-526 and those of this specification, the latter shall govern.

3.2 Material. The forceps shall be fabricated from Class 4, Type 410 or 420 material.

3.3 Hardness. The forceps, with the exception of the screw or pin, shall have a Rockwell hardness of C40 to C45 after proper stress relieving.

3.4 Style, design and dimensions. The forceps shall be of the Kelly design, suitable for accomplishing hemostasis in delicate tissue. The forceps shall be in accordance with Figure 1.

3.4.1 Locks. The forceps shall be of the box-lock type. The box-lock shall be accurately fitted and shall be without crevices, burrs, or sharp edges at any place in the lock construction. Pin or rivet shall be permanently secured. The joint performance shall be smooth, of equal resistance and non-binding in all positions when opening or closing the forceps (the lock may be lubricated). Further, under no circumstances shall looseness or freeplay of either shank permit any misalignment of the jaws in closing.

3.5 Finish. The finish shall be entirely mirror, satin, or matte. Should the instrument be of mirror finish, the locks and ratchets shall be polished smooth but need not possess a mirror finish. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure.

3.6 Identification marking. In addition to requirements of GG-I-526, forceps of foreign manufacture shall also be marked "stainless".

3.7 Jaw and ratchet operation. When tested as specified in 4.4.1, the jaw tips shall close first and progressively tighten with each ratchet position. Each ratchet position shall hold firmly and release instantly.

3.8 Workmanship. Workmanship shall be first class throughout. Forceps shall be free from defects which detract from their appearance or impair their serviceability.

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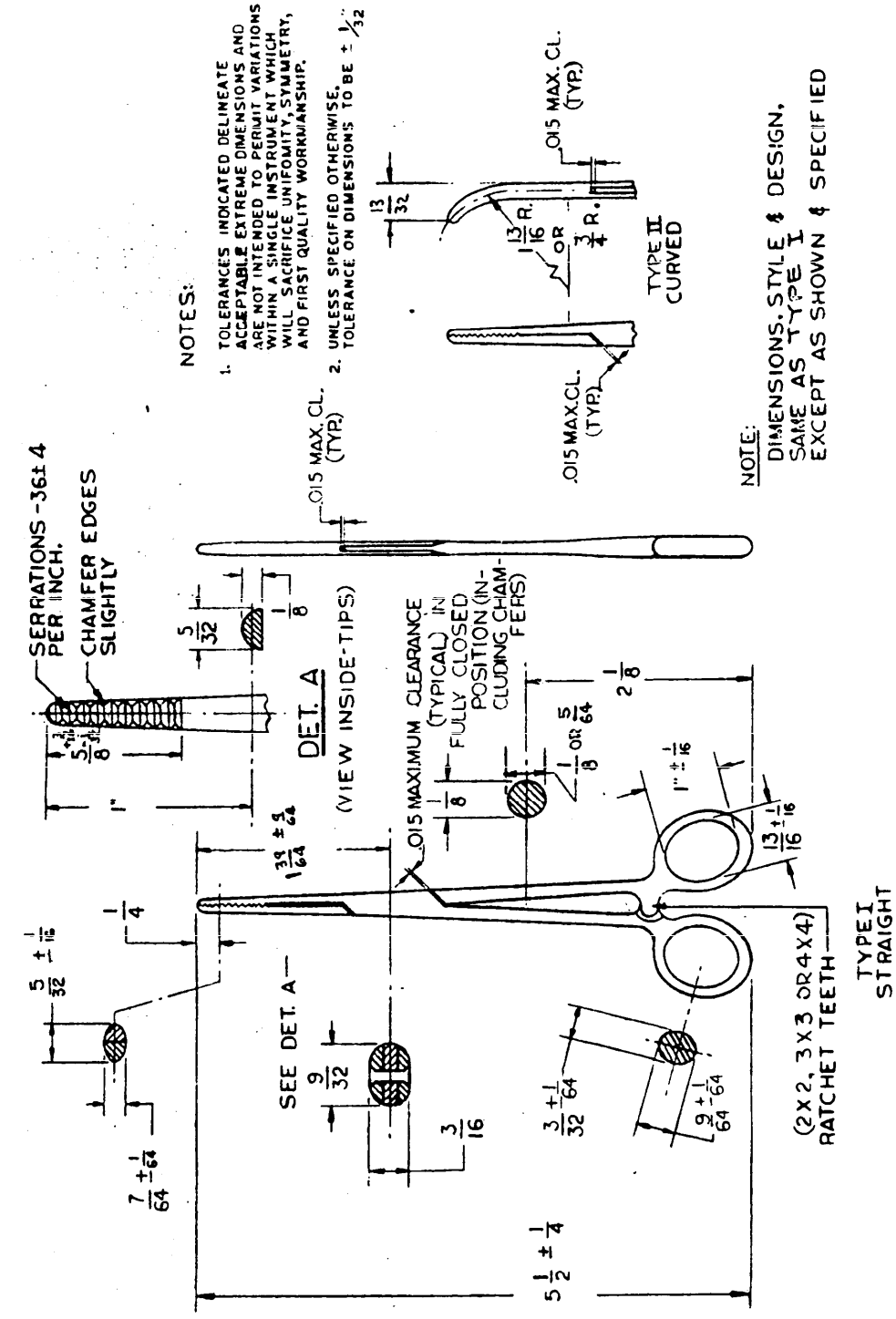


FIGURE 1. Forceps, hemostatic, Kelly, 5-1/2 inch.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the corrosion-resistant steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one hemostatic forceps.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (Percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Forceps	Hardness	3.3	4.4.2	S-2	1.5
	Dimensions	3.4 and GG-I-526	-------	S-2	2.5
	Boil test	GG-I-526	GG-I-526	S-2	1.0
	Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
	Distortion test	GG-I-526	GG-I-526	S-2	1.0
	Jaw and ratchet operation test	3.7	4.4.1	S-2	1.0

4.2.3 Raw materials. A sufficient amount from each lot of metal used in the construction of the forceps shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of forceps. The forceps shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination however, shall not be restricted to defects listed in Table III. Any forceps in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Style or design not as specified.
102	Forceps not free from cracks, pits, grind marks, dents, nicks, sharp edges, burrs, or overlaps.
103	Serrations not as specified.
104	Forceps not of box-lock design.
105	Forceps not free from excessive play in box-lock area.
106	Jaws do not mesh properly.
107	Serrations not free of foreign matter.
Minor	
201	Finish not as specified or not uniform.
202	Identification marking not present, correct, complete or legible.

4.3.2 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Jaw and ratchet operation test. The forceps shall be positioned to the first ratchet position, but not locked. Forceps with jaw tips not touching shall be rejected. Forceps having jaws which do not progressively tighten with each ratchet advance shall be rejected.

4.4.2 Hardness. Rockwell hardness shall be determined in acordance with ASTM E 18. A minimum of five hardness readings shall be made on test specimens representative of the material used in the fabrication of the instrument or on the finished instrument. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM E 140.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each forceps shall be cleaned and dried with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservatives shall not be applied.

5.1.1.3 Unit packaging. Each forceps shall be wrapped in neutral tissue paper or other suitable anticorrosive material. Each forceps shall be further packaged in accordance with MIL-P-116, method 1C-1.

5.1.1.4 Intermediate packaging. Twelve forceps shall be packaged in a box of appropriate size, constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the box specification.

5.1.2 Level C. Forceps shall be clean and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Three hundred and sixty forceps shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level a wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Three hundred and sixty forceps shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of box shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged forceps shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

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5.3 Marking. Each unit, intermediate package, and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The Kelly hemostatic forceps are to be used for accomplishing hemostasis in delicate tissue.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Type of forceps required (see 1.2).

6.3 NSN coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Type
6515-00-334-3800	FORCEPS, HEMOSTATIC, curved, Kelly, 5-1/2 inch	II
6515-00-334-6800	FORCEPS, HEMOSTATIC, Straight, Kelly, 5-1/2 inch	I

6.4 Matte and satin finish. It should be noted that matte finish specified herein can be distinguishable from mirror and satin finishes under high-power illuminated magnification. Forceps having matte finish shall have a reduced reflective surface that is more glare-resistant than satin finish.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD-MB

Preparing activity:
DLA-DM

Project No. 6515-2057

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4. TYPE OF ORGANIZATION *(Mark one)*

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